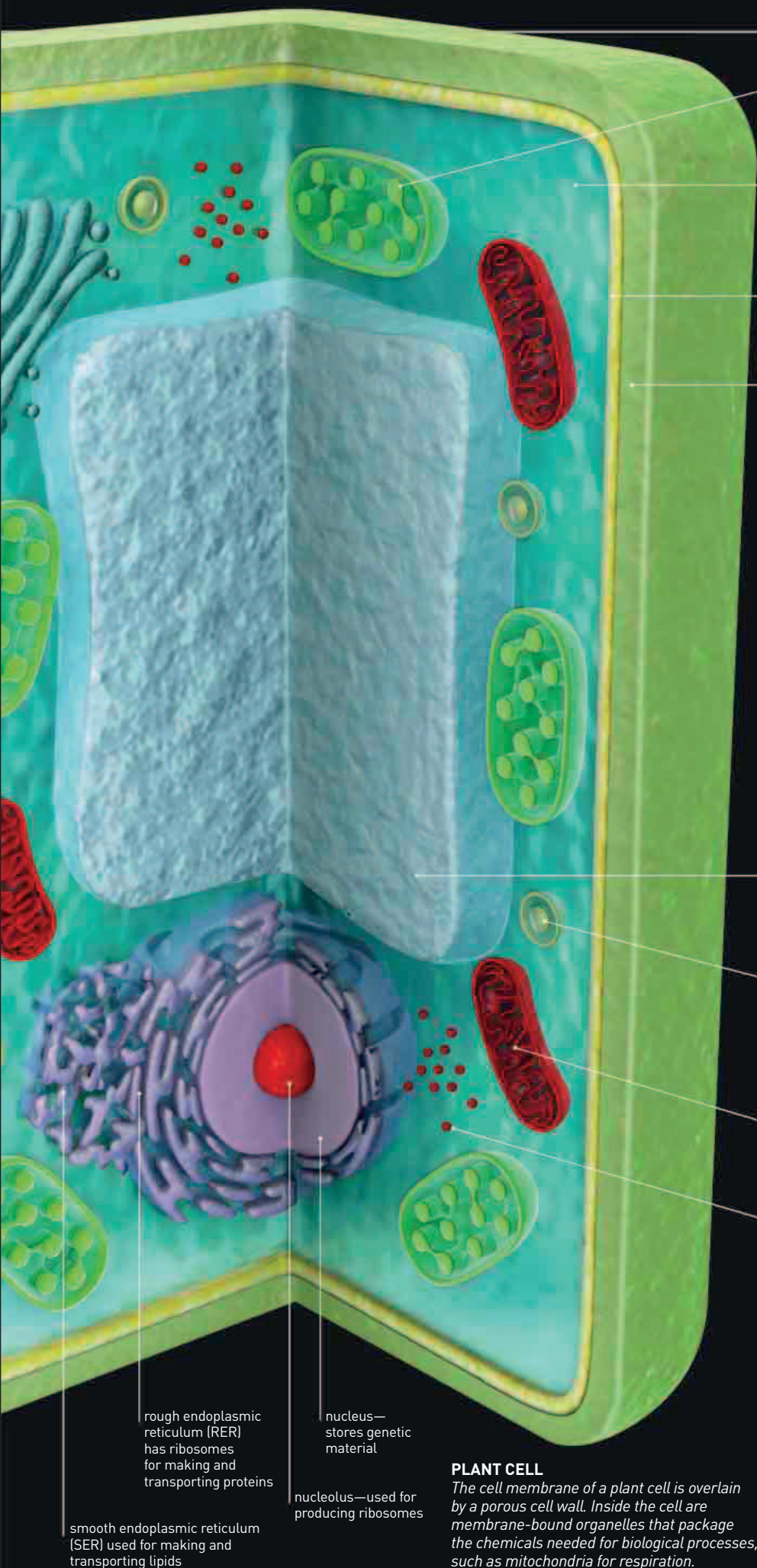




chloroplast—the site of food-producing reactions of photosynthesis

cytoplasm—fluid between the nucleus and the cell membrane

cell membrane—controls what enters and leaves the cell

cell wall—a rigid layer of cellulose fibers that supports the cell

vacuole—contains stored substances, pigment, or poisons

vesicle—a fluid-filled sac that stores cellular substances

mitochondrion—generates energy for the cell

ribosome—site where proteins are made

rough endoplasmic reticulum (RER) has ribosomes for making and transporting proteins

nucleus—stores genetic material

nucleolus—used for producing ribosomes

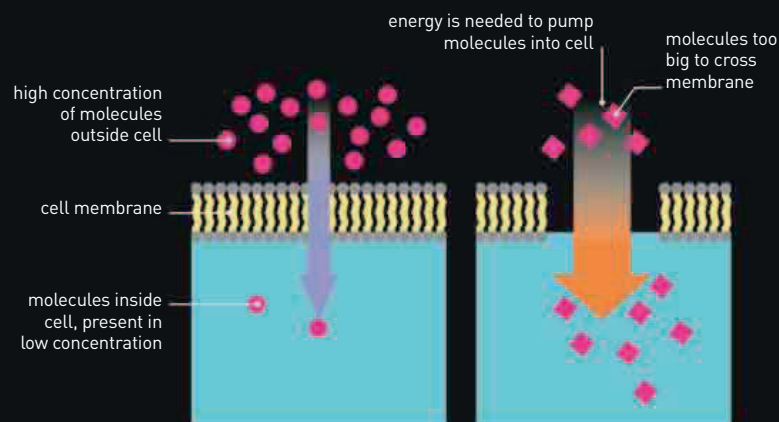
smooth endoplasmic reticulum (SER) used for making and transporting lipids

### PLANT CELL

The cell membrane of a plant cell is overlain by a porous cell wall. Inside the cell are membrane-bound organelles that package the chemicals needed for biological processes, such as mitochondria for respiration.

### MOVEMENT ACROSS CELL MEMBRANES

Cells and their organelles are bound by oily membranes that separate watery mixtures of chemicals on either side. Small molecules, or substances that can blend with oil, can penetrate the membrane—and tend to move from areas of high to low concentration by diffusion. Other particles can only get across by special molecular “pumps” in a process called active transport. The cell has to use up energy to do this. The energy comes from chemical processes of respiration (see below). Cells take up oxygen and excrete carbon dioxide by diffusion, but need active transport for moving salts and big molecules.



### DIFFUSION

This moves substances from high to low concentration. The bigger the difference (concentration “gradient”), the more they move.

### ACTIVE TRANSPORT

Here, substances move from low to high concentration, giving a cell the means to accumulate a substance in the cytoplasm or in an organelle.

### RELEASING ENERGY

Cells are driven by energy, which comes from food. Plants make their food by photosynthesis during which carbon dioxide and water react with light and chlorophyll. Almost all cells obtain energy in the same way: by breaking down a sugar called glucose. The process begins in the cell’s cytoplasm and finishes in organelles called mitochondria. These “powerhouses” use oxygen to extract energy in a particularly efficient way, generating a great deal of an energy-rich compound called ATP, which powers cellular activities involved in building or movement.



### MITOCHONDRION

Chemical reactions that release most of the cell’s energy happen on a mitochondrion’s internal membranes; the most active of cells have the most membrane-packed mitochondria.